

AIM

TA.110 SYSTEM CAPACITY PLAN

<Company Long Name>

<Subject>

Author: <Author>
Creation Date: June 14, 1999
Last Updated: August 15, 2025
Document Ref: <Document Reference Number>
Version: DRAFT 1A

Approvals:

<Approver 1>

<Approver 2>

Document Control

Change Record

Date	Author	Version	Change Reference
14-Jun-99	<Author>	Draft 1a	No Previous Document

Reviewers

Name	Position

Distribution

Copy No.	Name	Location
1	Library Master	Project Library
2		Project Manager
3		
4		

Note To Holders:

If you receive an electronic copy of this document and print it out, please write your name on the equivalent of the cover page, for document control purposes.

If you receive a hard copy of this document, please write your name on the front cover, for document control purposes.

Contents

Document Control	ii
Introduction	1
Purpose	1
Scope	1
Results and Conclusions	2
Capacity Planning Strategy	3
Capacity Plan Horizon	3
Business Volume Requirements	4
Production Environment Servers	5
Server Capacity Requirements	5
Processing Capacity Requirements - Server Class and CPUs	6
Memory Requirements	6
Disk Capacity Requirements	6
User Learning Environment Servers	8
Processing Capacity Requirements - Server Class and CPUs	8
Memory Requirements	8
Disk Capacity Requirements	8
Testing Environment Servers	10
Processing Capacity Requirements - Server Class and CPUs	10
Memory Requirements	10
Disk Capacity Requirements	10
Desktop Client Machines	12
Specifications	12
Processing Requirements	12
Network Capacity	13
Introduction	13
Bandwidth Requirements	13
Assumptions	15
<Assumption 1>	15
<Assumption 2>	15
Risks	16
<Risk 1>	16
<Risk 2>	16

Open and Closed Issues for this Deliverable	17
Open Issues.....	17
Closed Issues	17

Introduction

Purpose

The purpose of the **System Capacity Plan** document is to describe the system capacity requirements for the new systems and applications in <Company Short Name>'s architecture at production cutover and for some prescribed period of the future.

This document is intended to provide a reference at any time for resources working on the Application and Technical Architecture process of the <Project Name>. This document is also the source for current application and technical architecture information for project team members in processes with a dependency on the high level architecture model for <Company Short Name>'s future information systems.

The performance of the applications is directly related to the capacity of the machines that the applications are installed on. If there is insufficient processing capacity in the machines, the applications will perform poorly.

The task of planning capacity is designed to ensure that there is sufficient processing capacity to run the new Oracle Applications for the new system cut over time, and for some predetermined time into the future as the business of <Company Short Name> grows. If <Company Short Name> runs out of system processing capacity at some future time (either because of increased headcount, higher business volumes or both), the performance of the applications will start to suffer and <Company Short Name> may be faced with a decision to upgrade the application machines or move to a different more powerful class of machine to process the applications.

To ensure that the Oracle Applications implemented at <Company Short Name> will process the application load at cutover and for some period of time into the future, a capacity study has been performed. The method and results of the study are documented in this System Capacity Plan document.

Scope

This System Capacity Plan document covers the following data centers:

- <Data Center Name>
-
-

This System Capacity Plan document covers the following architecture components:

- Production Database Servers
- Production Application Servers
- Desktop Client Machines
- Print Servers
- Network Facilities

Results and Conclusions

The work on the system capacity plan has yielded the following results:

-
-
-

These results have the following consequences:

-
-
-

Capacity Planning Strategy

This section describes the strategy to be used for the system capacity planning exercise for the <Project Name> technical architecture.

Capacity Plan Horizon

The capacity planning horizon is the time period that you are capacity planning for after some initial date or event which is usually the date at which the new system cuts over into live production operations.

Capacity plan start date: <Live Production Date>

Capacity plan horizon: <Capacity Plan Horizon>

Business Volume Requirements

This section summarizes the business metrics to be used in capacity planning the system.

Production Environment Servers

This section describes the capacity planning of the servers that support the production environment. Capacity planning of the production servers are a critical part of helping to manage the performance quality of the technical architecture. It is necessary to reduce the risk of capacity shortfall, and any necessary re-architecture or migration to another server before production cutover or during the capacity plan horizon.

Server Capacity Requirements

Server Processing Capacity

Server Disk Capacity

Preliminary server sizing has been performed for the project servers using the Oracle Applications Sizing Spreadsheet. The results are summarized below with additional requirements for operating system and software tools. Note that these are servers to support the development of the production environments. The production environments are sized elsewhere.

The forms server and the database server or segregated for each environment. As modified source code, and application setup data will affect the purity of environments unless segregated.

Component	Concurrent Processing Server (MB)	Admin Server (Mb)	TRAIN Forms Server (Mb)	MAP Forms Server (Mb)	DEV Forms Server (Mb)	TRAIN Database Server (MB)	MAP Database Server (Mb)	DEV Database Server (Mb)
DNS Name	<Name>	<Name>	<Name>	<Name>	<Name>	<Name>	<Name>	<Name>
Operating System	0	0	0	0	0	0	0	0
Swap Space	0	0	0	0	0	0	0	0
User Directories	0	0	0	0	0	0	0	0
Documentation Repository	0	0	0	0	0	0	0	0
Oracle Executables	0	0	0	0	0	0	0	0
Application Executables	0	0	0	0	0	0	0	0
SYSTEM Tablespaces	n/a	n/a	n/a	n/a	n/a	0	0	0
Rollback Segments	n/a	n/a	n/a	n/a	n/a	0	0	0
Archiving files	n/a	n/a	n/a	n/a	n/a	0	0	0
Temp Space	0	0	0	0	0	0	0	0
Total Applications Data	n/a	n/a	n/a	n/a	n/a	0	0	0
Total Applications Indexes	n/a	n/a	n/a	n/a	n/a	0	0	0
Oracle Documentation Files	0	0	0	0	0	0	0	0
Space for Export files	n/a	0	n/a	n/a	n/a	0	0	0
Log files	0	0	0	0	0	0	0	0
Output files	0	0	0	0	0	0	0	0
TOTAL DISK SPACE	0	0	0	0	0	0	0	0

Processing Capacity Requirements - Server Class and CPUs

Memory Requirements

Disk Capacity Requirements

This section summarizes the disk space requirements for Oracle Applications and related tools to support <Company Short Name>'s production environment. Included below is a summary from the Oracle Applications spreadsheet. Software and data included in the sizing analysis are:

Oracle Applications™

- Application Object Library
- Oracle Alert®
- Oracle General Ledger
- Oracle Assets
- Oracle Receivables
- Oracle Payables
- Oracle Order Entry
- Oracle Purchasing
- Oracle Inventory
- Oracle Bills of Material
- Oracle Engineering
- Oracle Work in Process™
- Oracle Master Scheduling
- Oracle Capacity
- Oracle Cost Management
- Oracle Project Accounting
- Oracle Human Resources
- Oracle Payroll

Data Converted to Oracle Applications from Legacy Systems

-

-
-

Non-Oracle Applications

-
-
-

Development Tools and Other Software

-
-
-

Data Repositories

- Project Deliverable Library
-
-

The complete sizing details for each product are included below:

User Learning Environment Servers

This section describes the capacity planning of the servers that support the ongoing User Learning Environment. Capacity planning of the User Learning environment servers are a critical part of helping to manage the performance quality of the technical architecture. It is necessary to reduce the risk of capacity shortfall, and any necessary re-architecture or migration to another server before production cutover, or in the middle of the capacity plan horizon.

Processing Capacity Requirements - Server Class and CPUs

Memory Requirements

Disk Capacity Requirements

This section summarizes the disk space requirements for Oracle Applications and related tools to support <Company Short Name>'s environments on the <Project Database Server Name>. Included below is a summary from the Oracle Applications sizing spreadsheet. Software and data included in the sizing analysis are:

Oracle Applications

- Application Object Library
- Oracle Alert
- Oracle General Ledger
- Oracle Assets
- Oracle Receivables
- Oracle Payables
- Oracle Order Entry
- Oracle Purchasing
- Oracle Inventory
- Oracle Bills of Material
- Oracle Engineering
- Oracle Work in Process
- Oracle Master Scheduling
- Oracle Capacity
- Oracle Cost Management
- Oracle Project Accounting

- Oracle Human Resources
- Oracle Payroll

Data Conversion Requirements

-
-
-

Development Tools and Other Software

-
-
-

Project Documentation Library

-
-
-

Testing Environment Servers

This section describes the capacity planning of the servers that support the ongoing Testing environment. Capacity planning of the testing servers are a critical part of helping to manage the performance quality of the technical architecture. It is necessary to reduce the risk of capacity shortfall, and any necessary re-architecture or migration to another server before production cutover, or in the middle of the capacity plan horizon.

Processing Capacity Requirements - Server Class and CPUs

Memory Requirements

Disk Capacity Requirements

This section summarizes the disk space requirements for Oracle Applications and related tools to support <Company Short Name>'s environments on the <Project Database Server Name>. Included below is a summary from the Oracle Applications sizing spreadsheet. Software and data included in the sizing analysis are:

Oracle Applications

- Application Object Library
- Oracle Alert
- Oracle General Ledger
- Oracle Assets
- Oracle Receivables
- Oracle Payables
- Oracle Order Entry
- Oracle Purchasing
- Oracle Inventory
- Oracle Bills of Material
- Oracle Engineering
- Oracle Work in Process
- Oracle Master Scheduling
- Oracle Capacity
- Oracle Cost Management
- Oracle Project Accounting

- Oracle Human Resources
- Oracle Payroll

Data Conversion Requirements

-
-
-

Development Tools and Other Software

-
-
-

Project Documentation Library

-
-
-

Desktop Client Machines

Specifications

The following table lists the specifications for the minimum recommended client processing capacity to run Oracle Applications:

Component	Desktop User	Laptop User	Developer
CPU	Pentium	Pentium	Pentium
Clock Speed	120 MHz	120 MHz	200 MHz
RAM	16 MB	16 MB	32 MB
Disk	200 MB		1 GB
Monitor/Display	1024 x 768 SVGA	1024 x 768 SVGA	1024 x 768 SVGA

Processing Requirements

The different user roles in <Company Short Name>'s business will have different client machine processing requirements. This section discusses the individual role requirements for the client desktop machines they will use.

Client Machine Resource	<User Role 1>	<User Role 2>			
Machine Class	Desktop	Laptop			
Vendor	Dell	Toshiba			
CPU	Pentium P5	Pentium P5			
Clock Speed	150Mhz	150Mhz			
RAM	16 MB	32 MB			
Disk	810 MB	1.2 GB			
Monitor	1024 x 768 SVGA	Laptop 640x480 VGA			
Operating System	Win 3000	Win 3000			
Network Interface Card					

Network Capacity

Introduction

The performance of an application that runs across a network is very dependent upon the performance of the network that the application runs over. Oracle Applications technical architecture requires communications between several tiers of the architecture through a network connection. If the network between the client and the application or database server is performing poorly, the performance of the applications will necessarily be affected, even if the client and database server are high performance machines.

The two basic factors affecting network performance are network latency and network capacity. The network latency is not an additive performance parameter like network user bandwidth, but a large volume of user traffic can affect the measured network latency time. Latency is intrinsically related to the network design including the number of network "hops" that connect two locations.

Bandwidth Requirements

Oracle Applications Requirements

The following table summarizes the bandwidth requirements for the users in each site, across each network segment that connects the sites of the client to the database server site.

Site 1 - Client	Business Function	No. of Users	Site 2 - Server	Network Type	Network Segment	Bandwidth Kbps/user	Comments

[illegible]

Assumptions

The following assumptions were made in creating this System Capacity Plan:

1. <Assumption 1>
2. <Assumption 2>

These are individually described in more detail below.

<Assumption 1>

<Assumption 2>

Risks

The System Capacity Plan presented here carries the following risks:

1. <Risk 1>
2. <Risk 2>

These are individually described in more detail below.

<Risk 1>

<Risk 2>

Open and Closed Issues for this Deliverable

Open Issues

ID	Issue	Resolution	Responsibility	Target Date	Impact Date

Closed Issues

ID	Issue	Resolution	Responsibility	Target Date	Impact Date